

VIGNESHWARI JAYAPRAKASH

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Machine Learning · Statistical Modeling · Experimentation · MLOps · Data Engineering

SUMMARY

Data Scientist with **5+ years** productionalizing ML across financial services and enterprise. Shipped fraud detection on **1M+ transactions/day** at **less than 200ms latency**, generating **\$1M+ annual savings**. Strong in Python, SQL, A/B testing, causal inference, and end-to-end MLOps.

EDUCATION

M.S. Data Science — Arizona State University, Tempe, AZ

Aug 2024 – May 2026

GPA: 4.0/4.0 · Computing & Decision Analytics · ASU Scholar

B.Tech Information Technology — Anna University, India

Jun 2009 – Apr 2013

GPA: 3.6/4.0 · Gold Medalist

PROFESSIONAL EXPERIENCE

Data Scientist Intern — New Mexico Department of Information Technology

Jun 2025 – Aug 2025

- Deployed XGBoost + Isolation Forest anomaly detection via FastAPI on AWS, lifting detection from **62% (rule-based)** to **85%** and cutting false positives by **60%** on 10K+ daily logs.
- Engineered 14 features from incident logs via SQL ETL pipelines, raising F1 from **0.71** → **0.87** (+22%).
- Built KPI Power BI dashboard adopted by state leadership, earning **CIO recognition** for operational impact.

Technology Analyst – Data Science & Analytics — Infosys (Client: BNSF Railway)

Oct 2017 – Jan 2019

- Productionalized Python fraud detection on **1M+ transactions/day** at **less than 200ms p95 latency**, generating **\$1M+ annual savings**; recognized by **CFO of BNSF**.
- Built PySpark + Databricks time-series forecasting (ARIMA, Prophet) for **500+ railway assets**; MAPE **8.4%**, eliminating **\$400K/yr** in unplanned downtime.

Software Engineer – Machine Learning — Infosys (Client: BNSF Railway)

Oct 2013 – Sep 2017

- Designed A/B framework for vendor segmentation across **50K+ vendors**, reducing manual review by **30%** (~600 analyst hrs/quarter) at $p < 0.01$.
- Led migration of sourcing platform to SAP and replaced Mainframe reports with Tableau dashboards, cutting reporting time by ~50% and reducing operational costs by ~70%.
- Mentored a 10-person team on ML workflows; promoted twice in 3 years; **Insta Award** winner.

ADDITIONAL TRAINING

Self-directed upskilling in LLM Architectures, RAG Systems, and Cloud-Native ML (2019–2024)

SELECTED PROJECTS

SentinelRAG — NLP Compliance Screening & Audit System

2026

- RAG pipeline (LangChain + ChromaDB) screening **18,700+** compliance entities at **99% recall / 100% precision** with deterministic audit trail, replacing manual review of ~400 cases/week.
- Optimized FastAPI inference on AWS via batching + semantic caching: latency **–40%** (820ms → 490ms), throughput **3×** (45 → 135 RPS).
- Technologies: Python, LangChain, ChromaDB, FastAPI, Docker, AWS, Streamlit*

E-Commerce Hybrid Recommender (Two-Tower + LightGBM Ranker)

2025

- Two-Tower neural retriever with FAISS ANN over **2M+ products**; cut inference latency **60%** (180ms → 72ms) at constant recall@100.
- LightGBM ranker on weighted conversion objectives tracked in MLflow: MRR **+22%**, NDCG@10 **+18%** vs. baseline.
- Technologies: Python, TensorFlow Recommenders, LightGBM, FAISS, Sentence Transformers, MLflow, FastAPI*

TECHNICAL SKILLS

Languages	Python, SQL, R, PySpark, Shell
ML / Stats	PyTorch, TensorFlow, Keras, scikit-learn, A/B Testing, Hypothesis Testing, Causal Inference
LLM / NLP	LLMs, RAG, LangChain, Prompt Engineering, Vector Search
Data Eng	Databricks, ETL, Pandas, NumPy
MLOps & Cloud	MLflow, Docker, CI/CD, AWS (S3, Lambda, EC2), Git
Visualization	Tableau, Power BI, Matplotlib, Seaborn
APIs & Vector	FastAPI, REST, Streamlit, FAISS, ChromaDB